

Speech Emotion Recognition

From Raw Voice to Emotions

An audio intelligence pipeline

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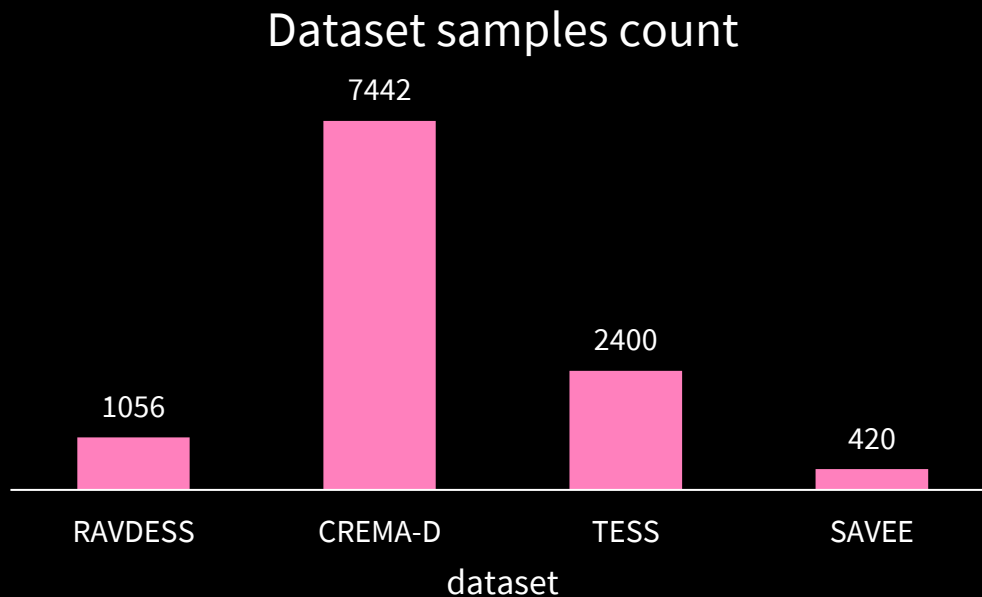
Why is it Important?

- Emotion-Aware Voice Assistants
 - Google Assistant, Siri, Alexa
 - Adjust responses based on detected emotion
- Mental Health Support & Monitoring
 - Detect early signs of depression, anxiety
 - Daily voice check-ins in wellness apps
- Call Center Analytics
 - Real-time emotion detection during customer support
 - Prioritize angry/frustrated customers



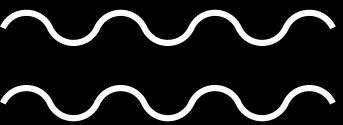
Dataset Overview

- Contains **4 individual dataset**: CREMA-D, SAVEE, TESS, RAVDESS
- **6 Emotions**: Angry, Happy, Sad, Fear, Disgust, Neutral



Total audio
samples

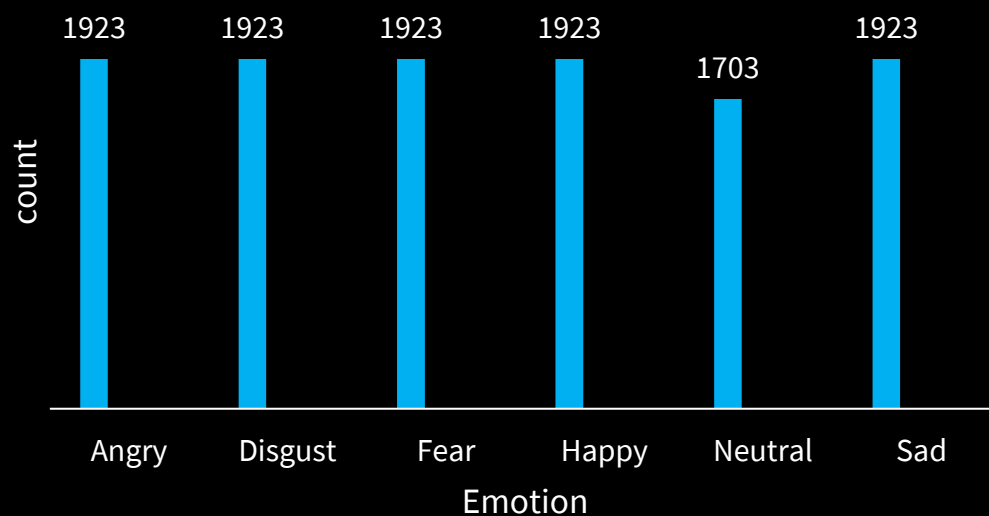
11,318



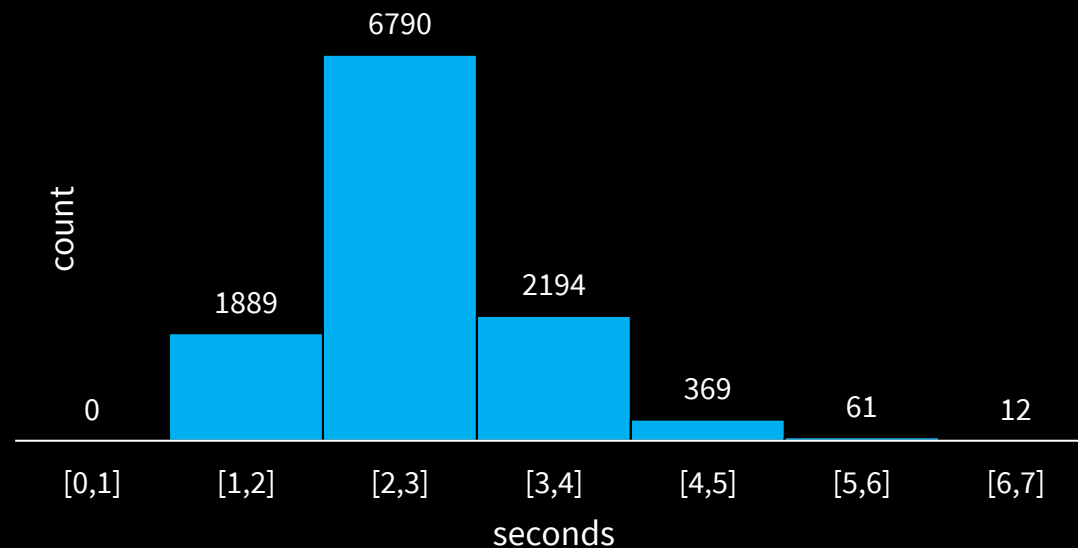
Dataset Overview

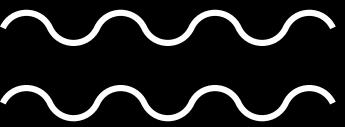
Duration
Average
2.54 s

Distribution of audios by emotions

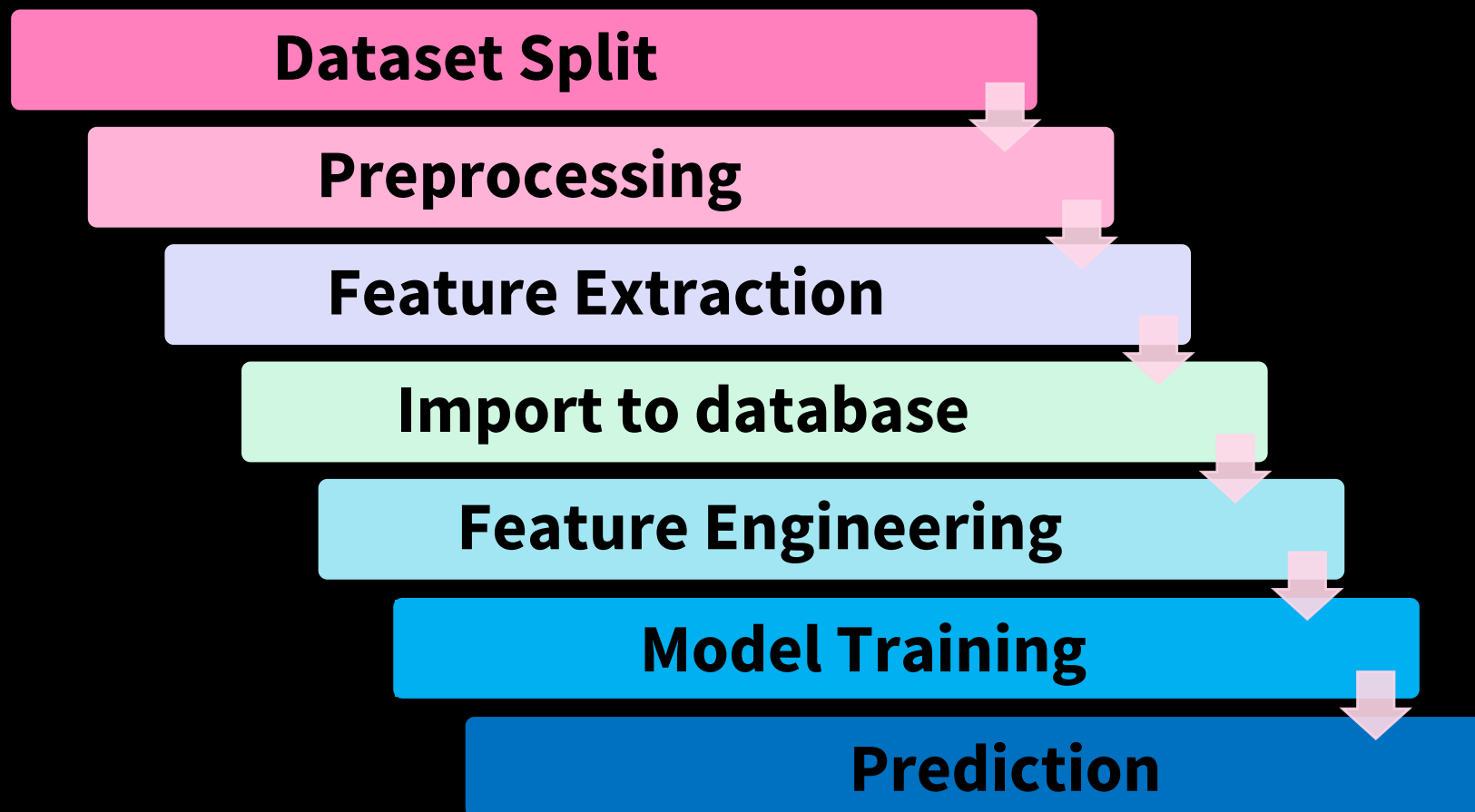
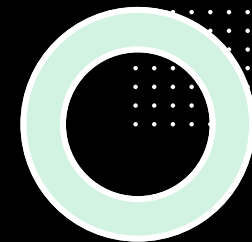


Distribution of audios by duration

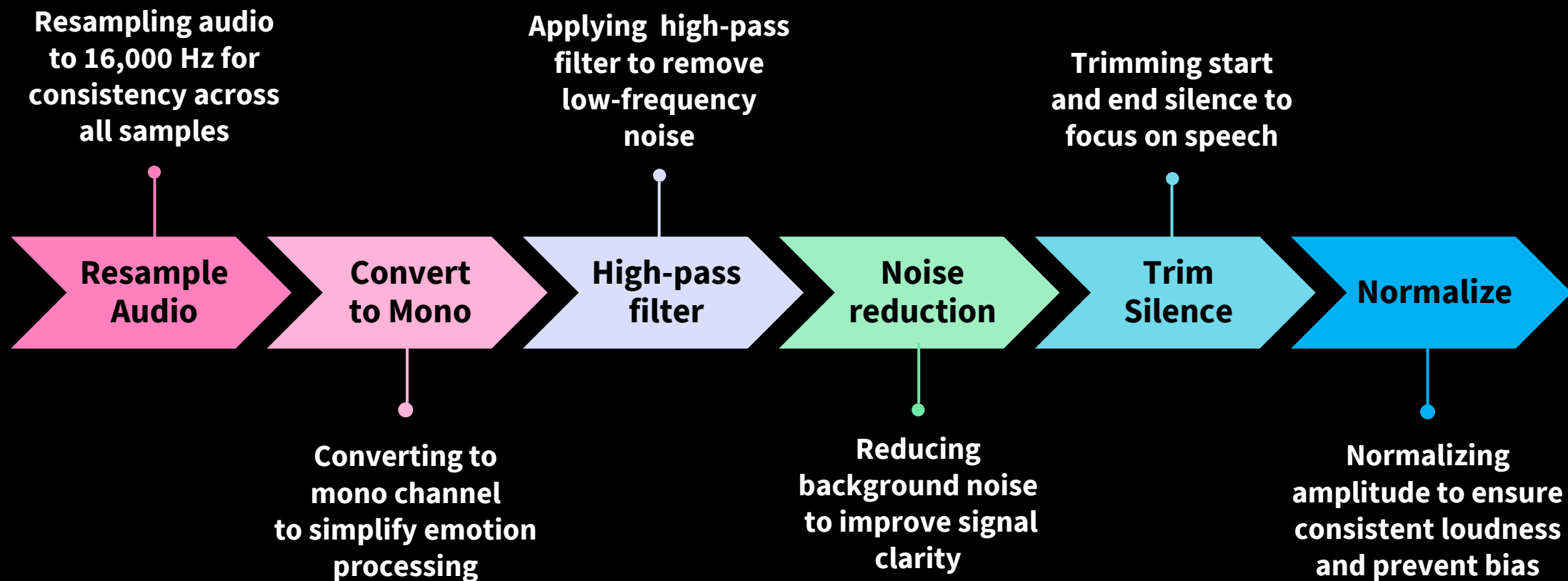




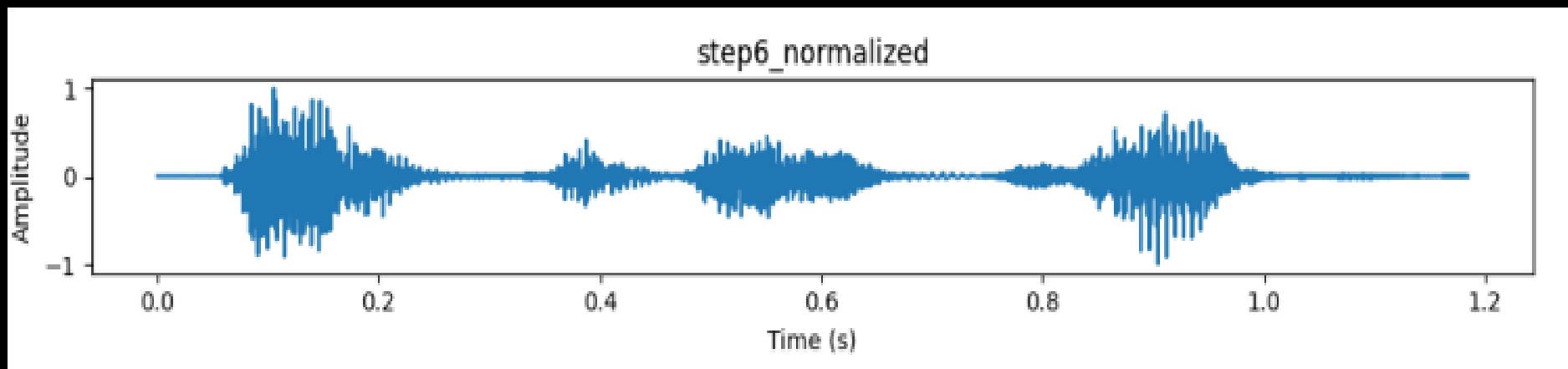
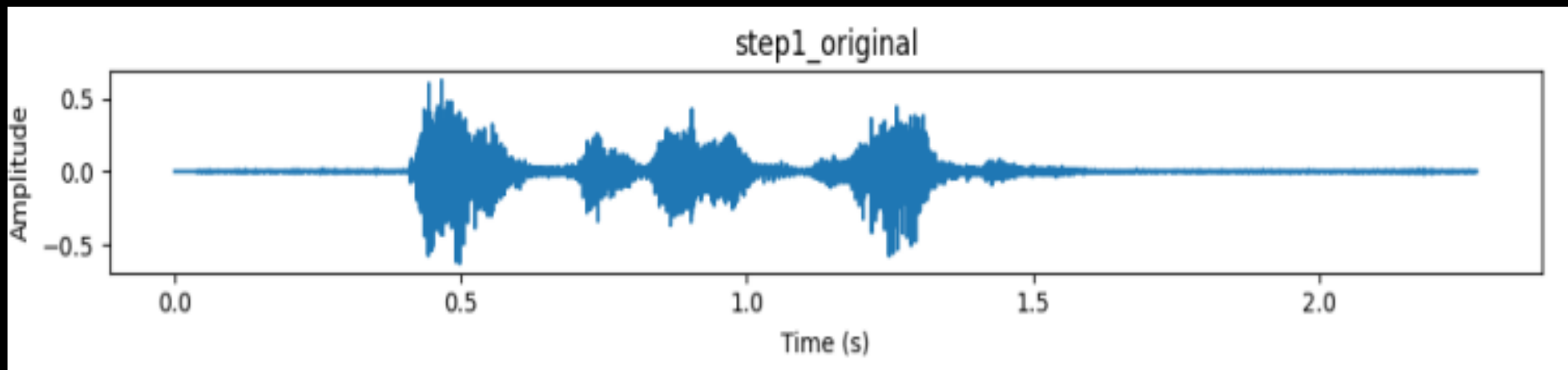
Pipeline Flow



Preprocess Flow



Before and After Preprocess



Approach 1 - Splitting Audio into Chunks to Ensure Uniform Audio Length



We split each audio into **0.8s** chunks
with **0.4s** overlap.

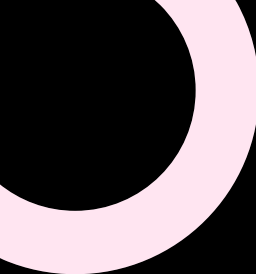
Overlap is to preserve emotional cues
that span across word boundaries.

But why 0.8s of Duration?

Each word
takes on
average

0,4))
seconds
to speak





Feature Extraction Strategy



We extract features frame-by-frame using a sliding window with an overlap.

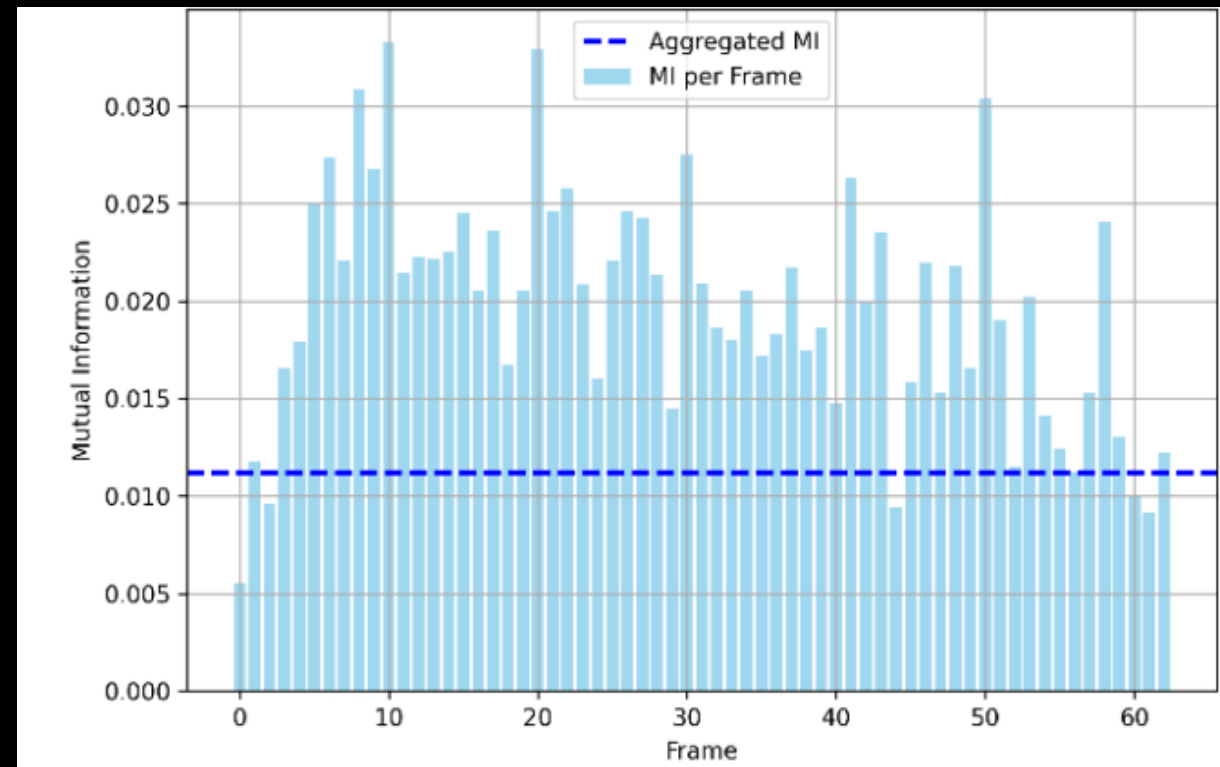
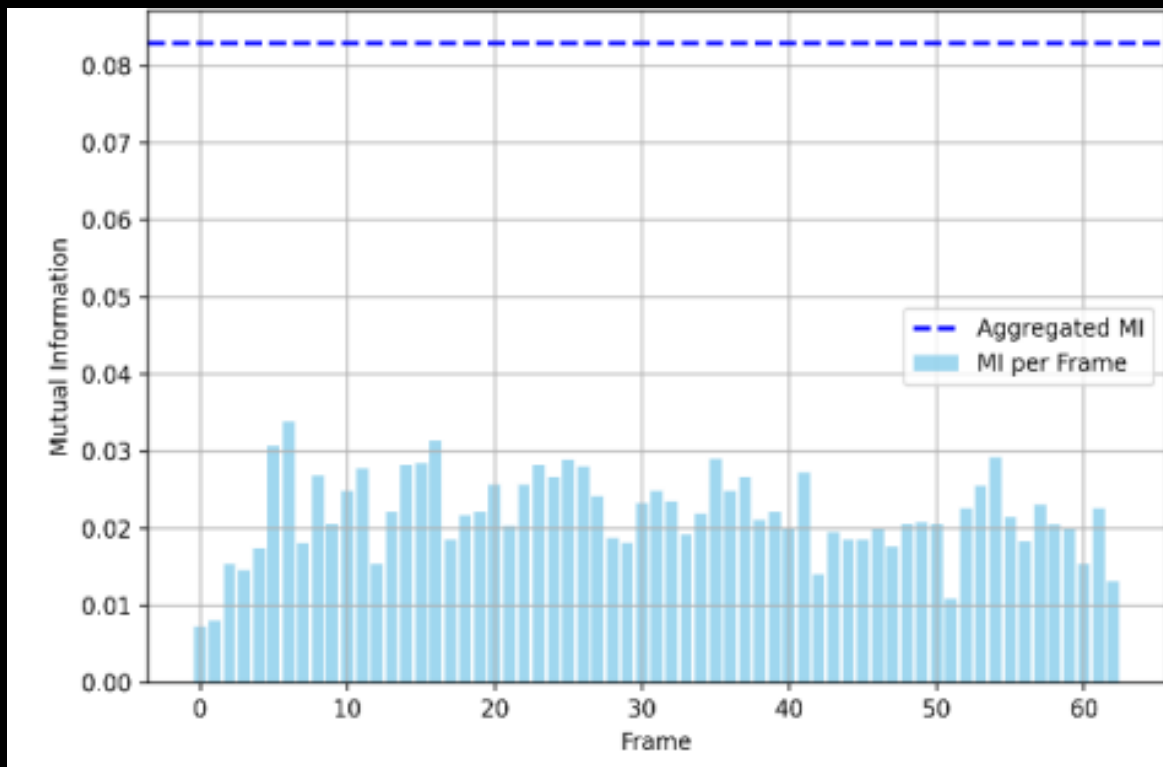
Frames per 0.8s chunk: 63 frames

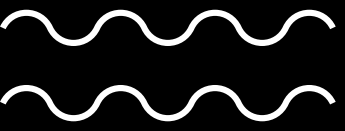


Aggregated or Framewise?

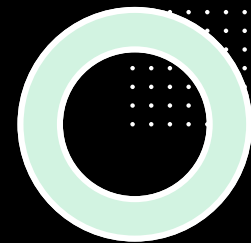


Kept frame wised if there is at least one frame that has MI more than aggregated MI, otherwise the aggregated version is kept.





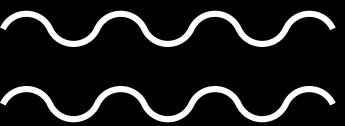
Challenges with approach 1



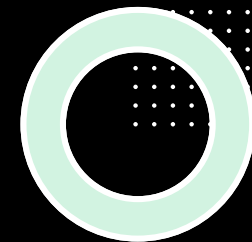
Keeping some features frame-wise significantly
Increased the dimensionality.
With limited data, the model struggled to learn effectively,
resulting in poor performance.



What if we keep all the features aggregated?

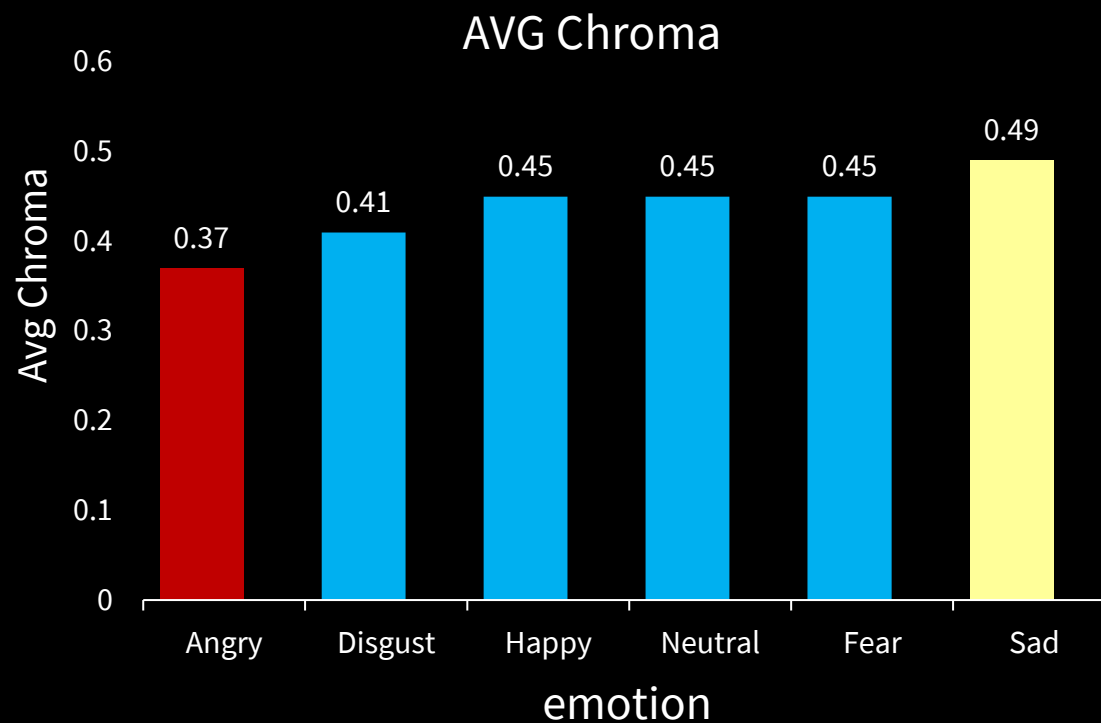
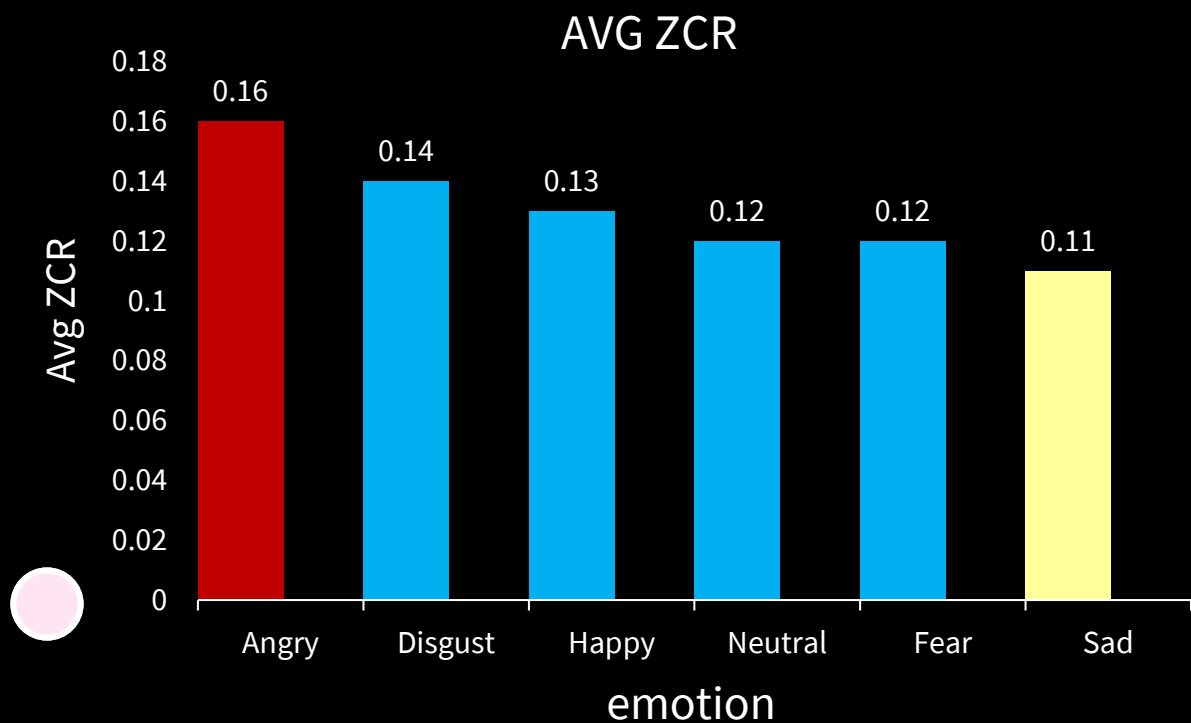


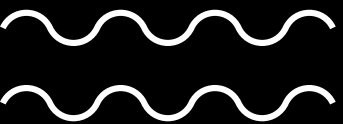
Chroma vs ZCR Feature



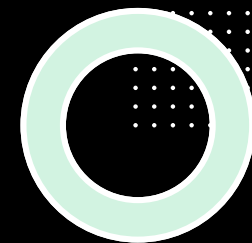
ZCR (zero crossing rate) is the number of times the waveform crosses the horizontal axis.

Chroma captures harmonic and melodic characteristics of audio.

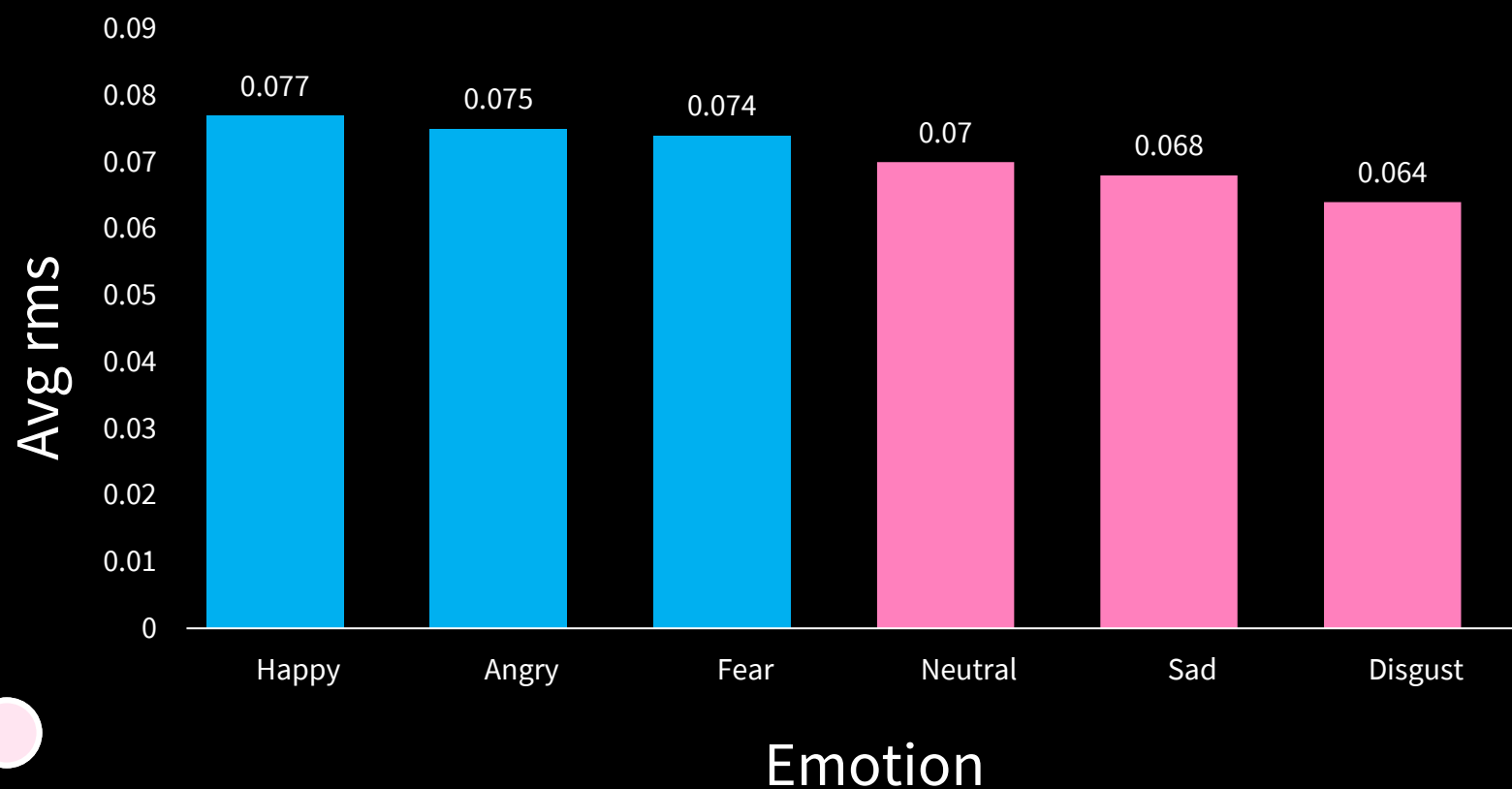




RMS Feature

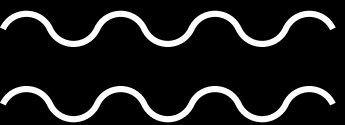


AVG RMS

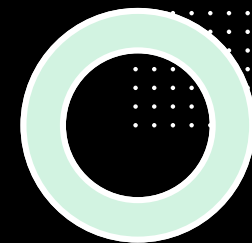


Rms captures the energy or loudness of a signal. This feature seems to be an effective feature for distinguishing high-intensity emotions from lower-intensity emotions.





MFCC Feature



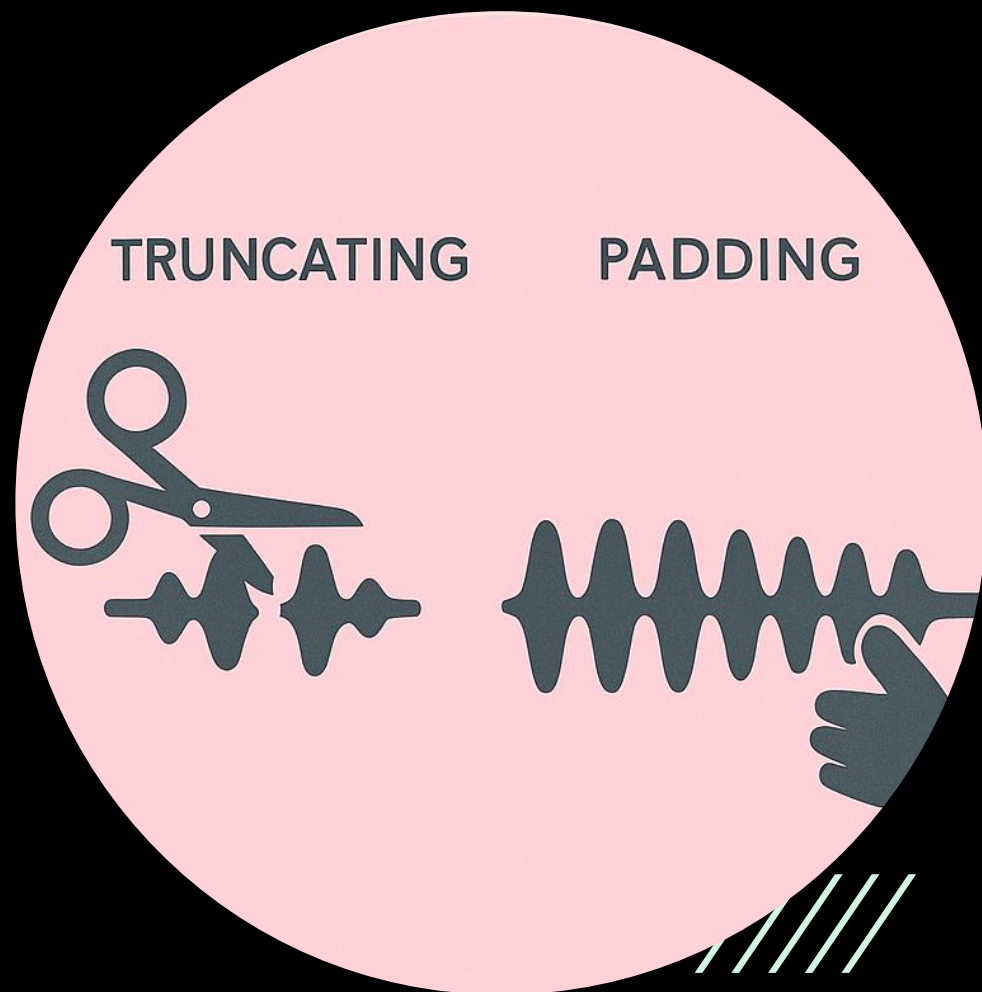
MFCCs are numbers that describe how voice energy is spread across frequencies, creating a compact fingerprint of how the voice sounds based on human hearing.

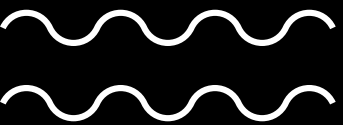


Approach 2 - Aggregating all features

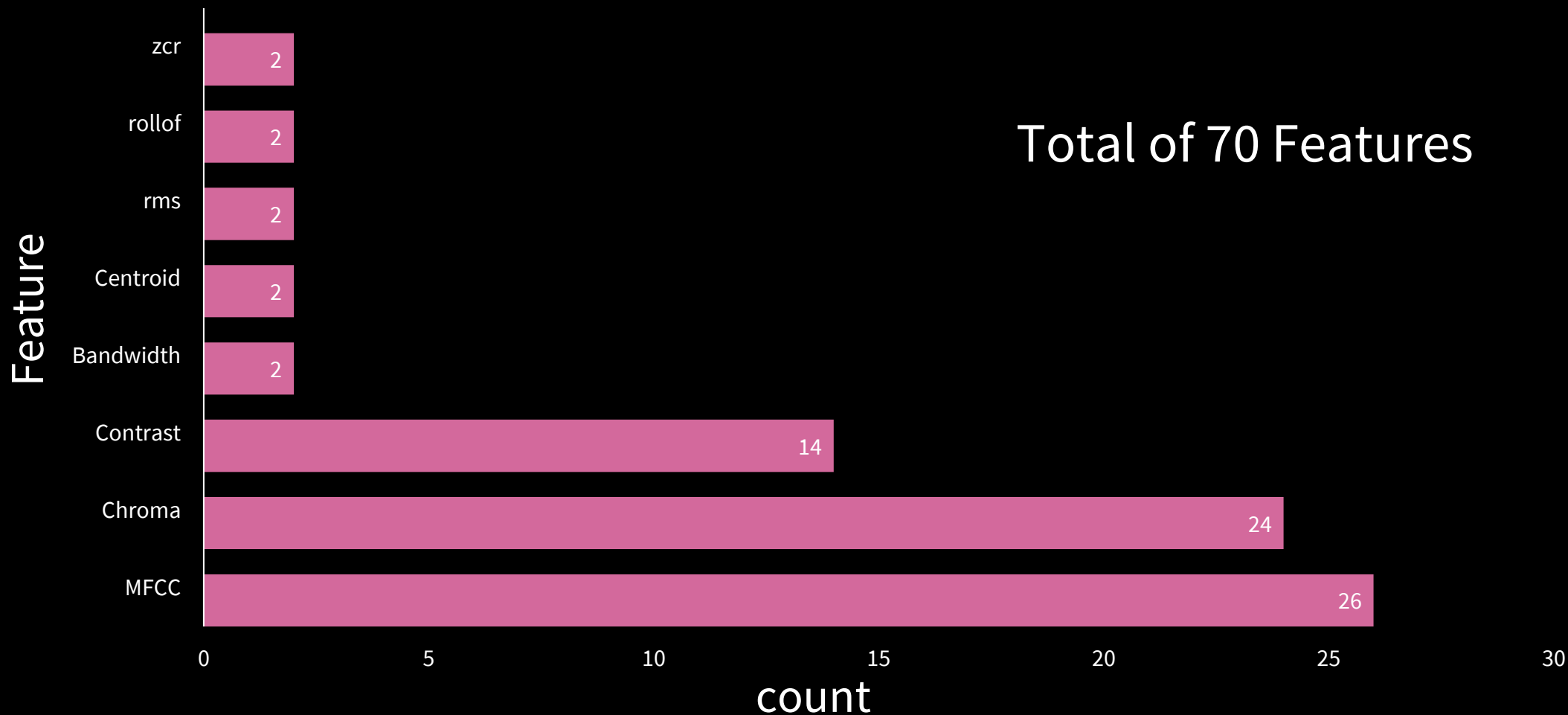
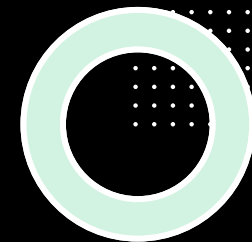


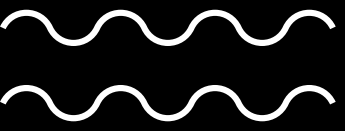
The average duration of audios were 2.5s. When loading the audios we just kept the first 2.5s with an offset 0.6s . Shorter audio clips are padded, while longer ones are truncated.



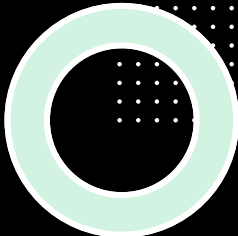


Features categories





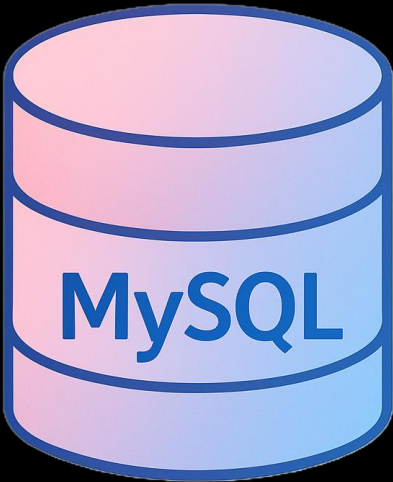
Database

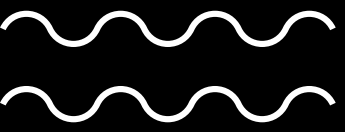


FEATURES TABLE	
♀	file name
	emotion
	features (70)

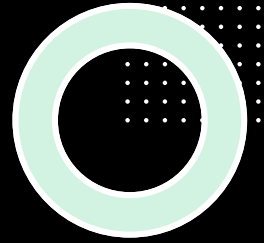


META DATA TABLE	
♀	file name
	actor
	sex
	sentence
	duration

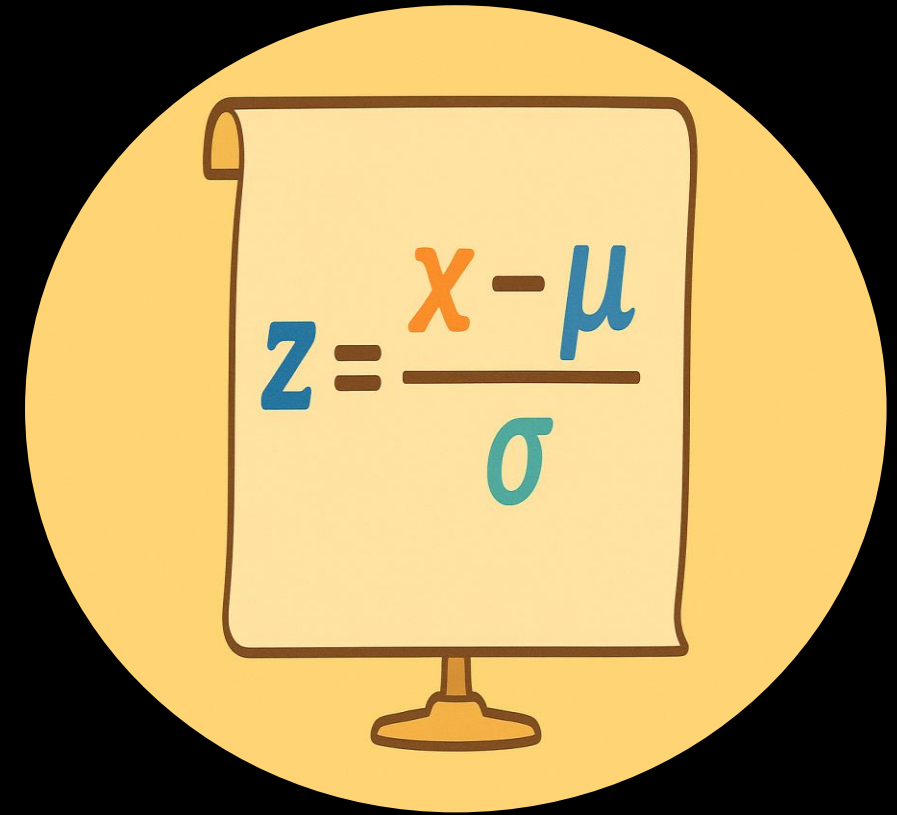


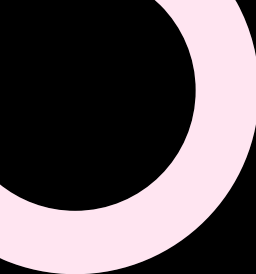


Feature Engineering



1. Standard Scaling :
fit on training data and transform
it on validation and test.
2. Emotion Encoding:
fit the encoder on the training data
and transform it on validation and test.





Training Models



Traditional Machine Learning Models:

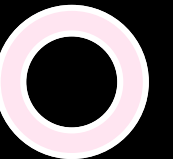
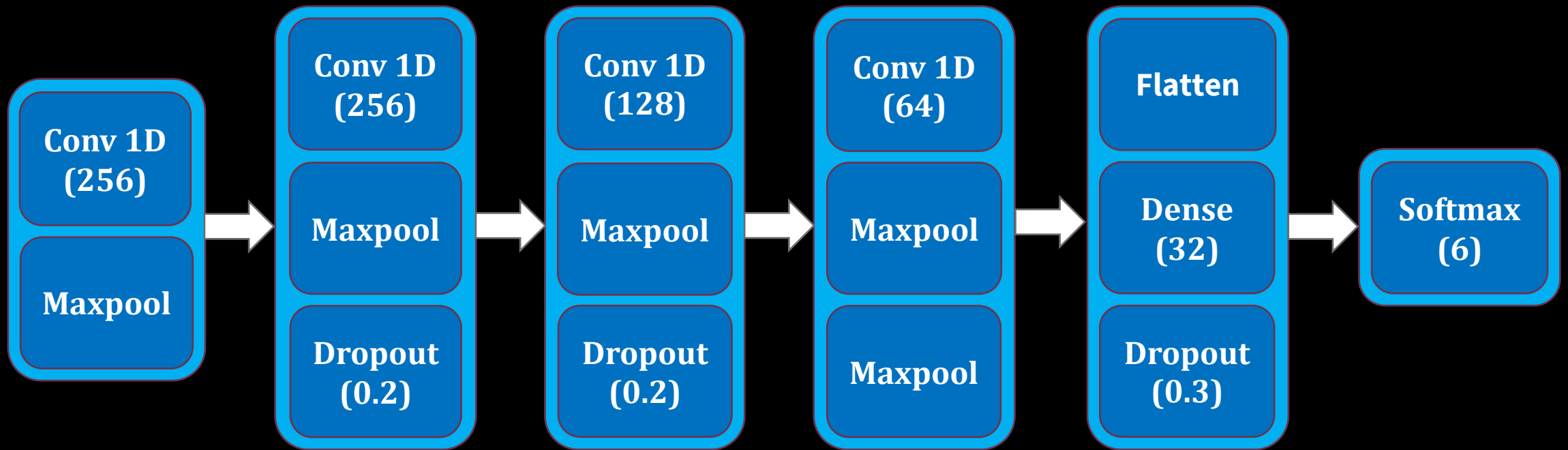
- XGBoost
- SVM (Support Vector Machine)
- Random Forest

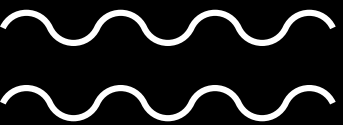
Neural Network / Deep Learning Models:

- CNN (Convolutional Neural Network)
- MLP (Multi-Layer Perceptron)



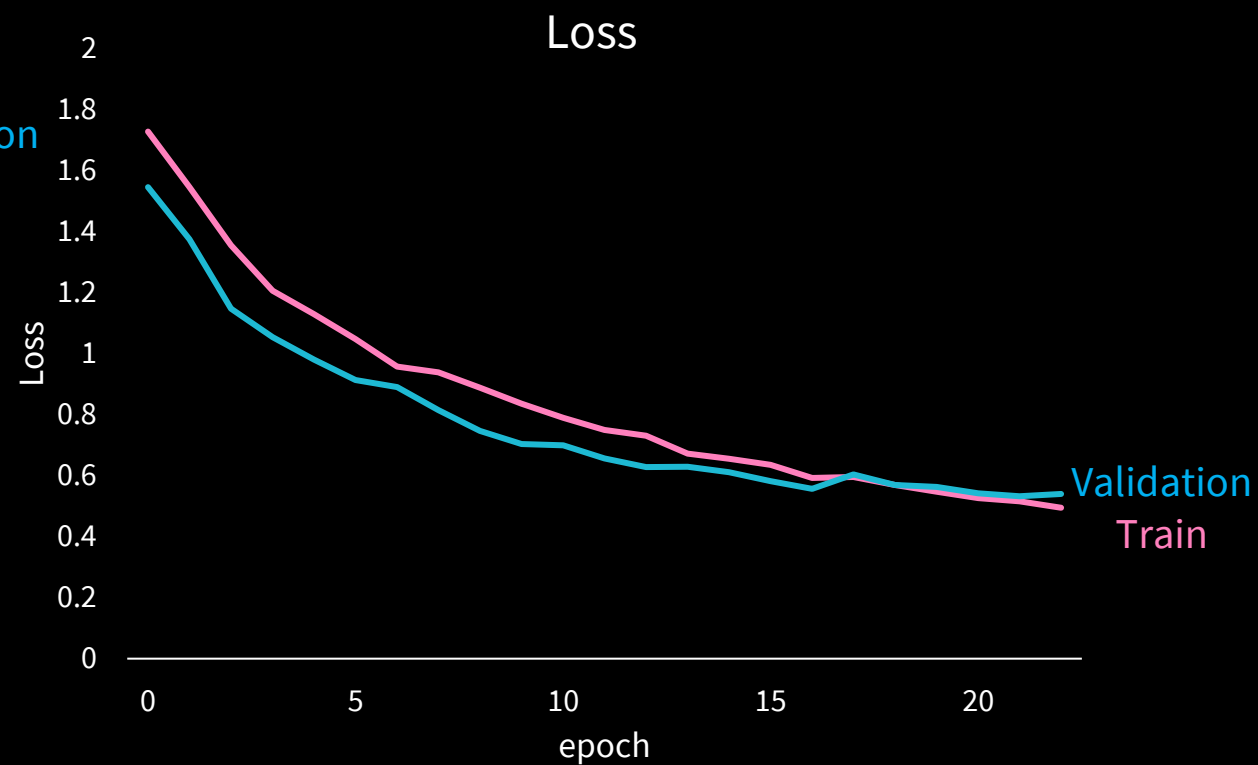
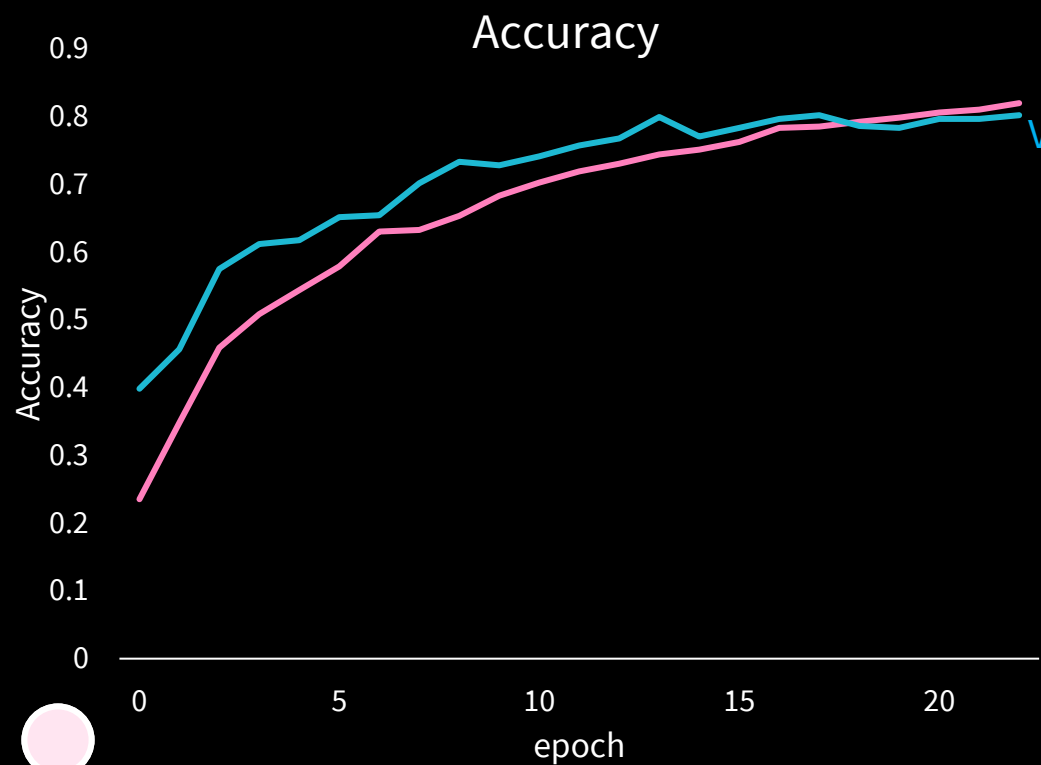
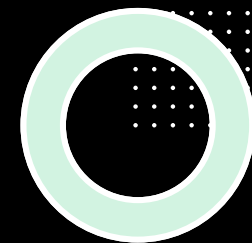
CNN Architecture

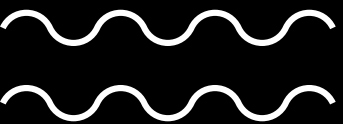




CNN

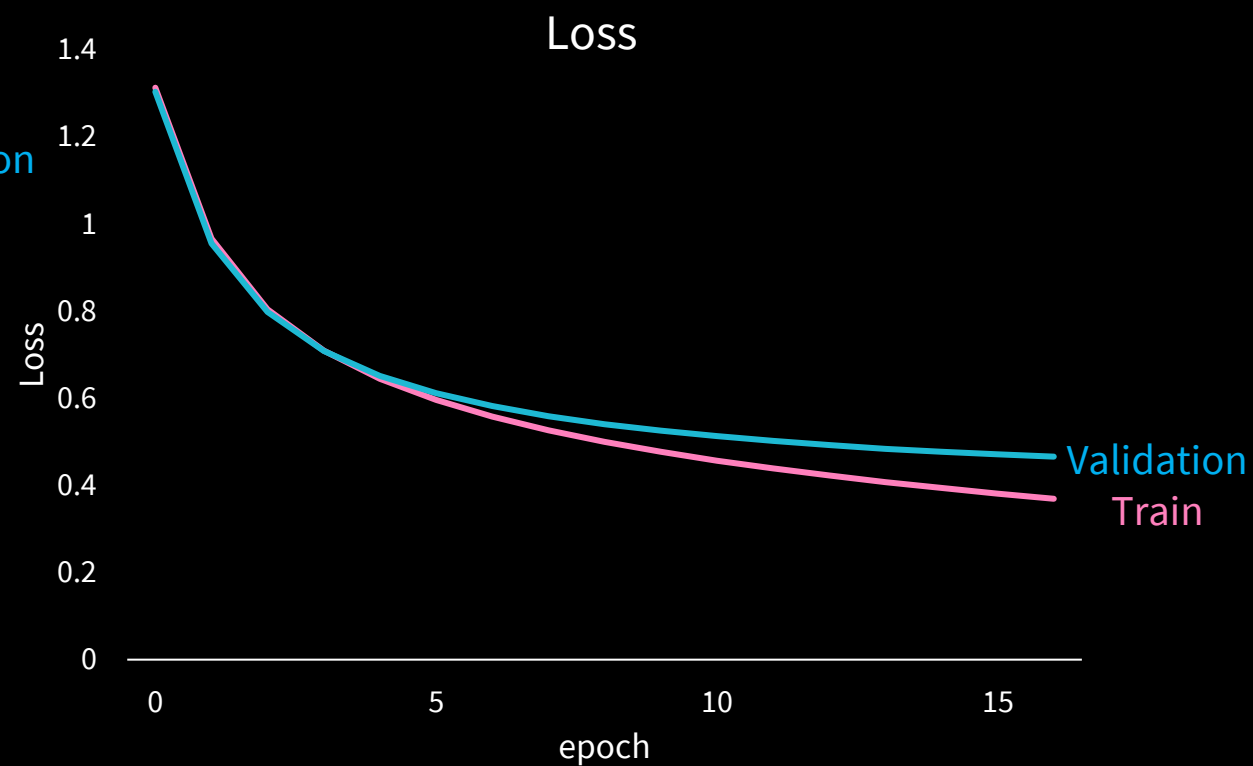
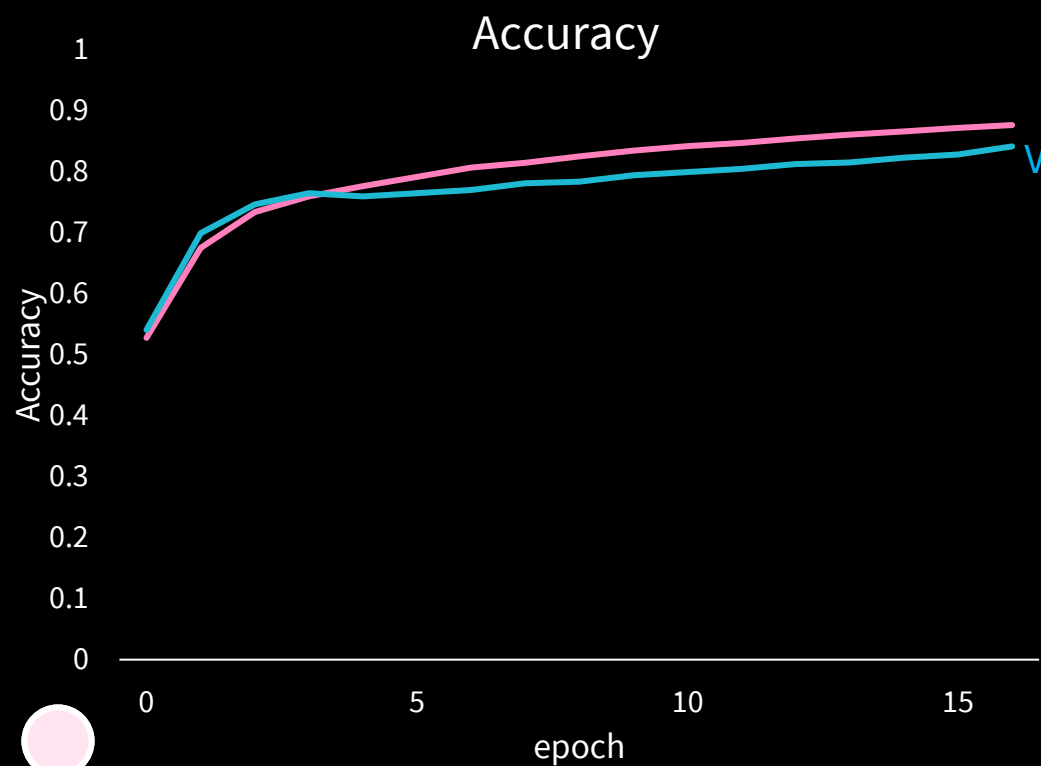
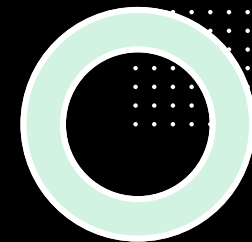
Test Accuracy = 77.8%
Test F1 score = 77.9%

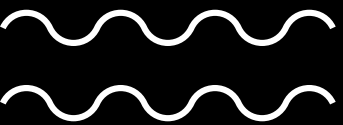




MLP

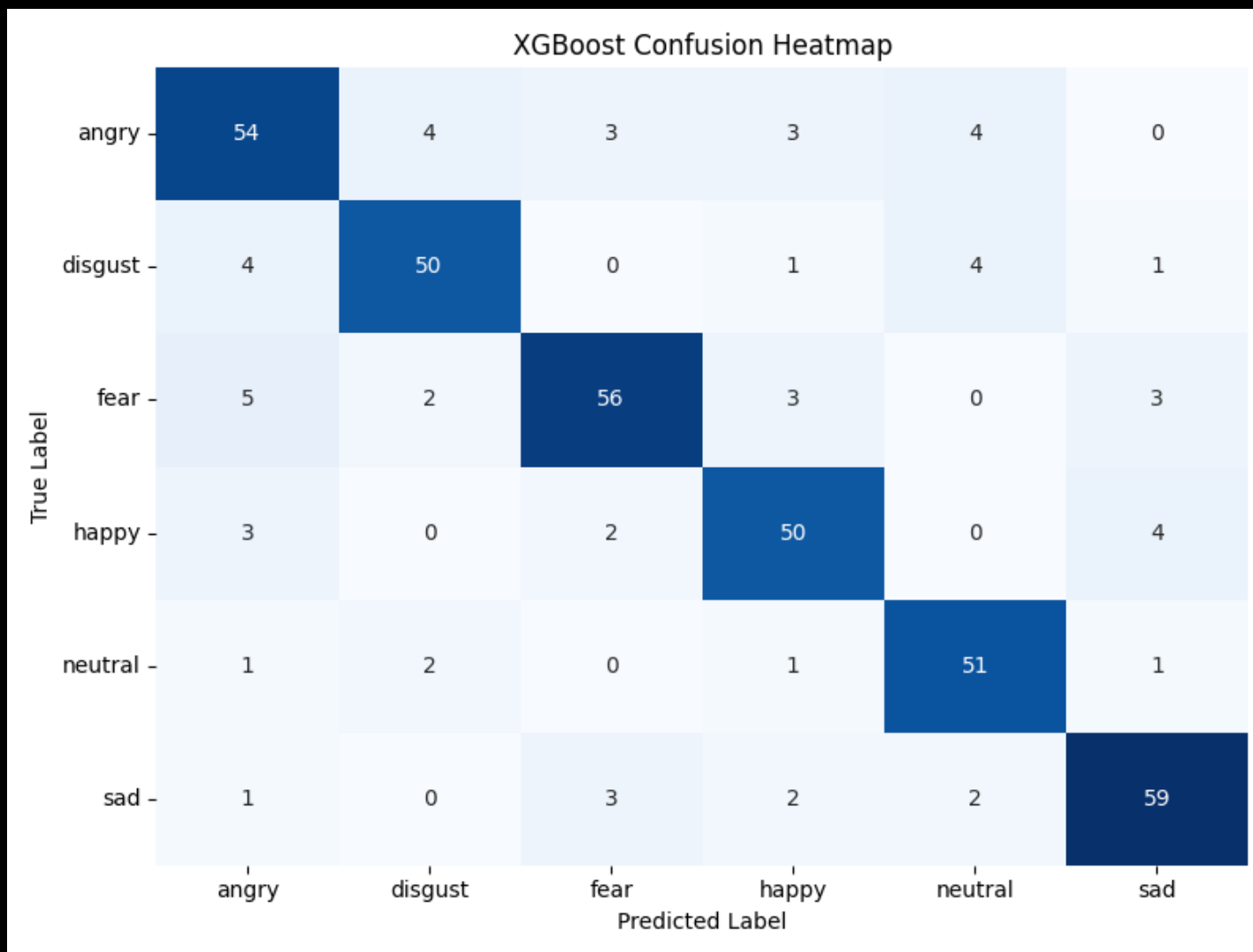
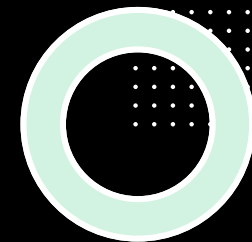
Test Accuracy = 85.6%
Test F1 score = 85.59%

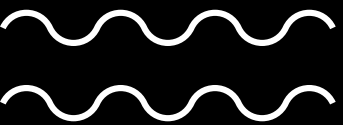




XGboost

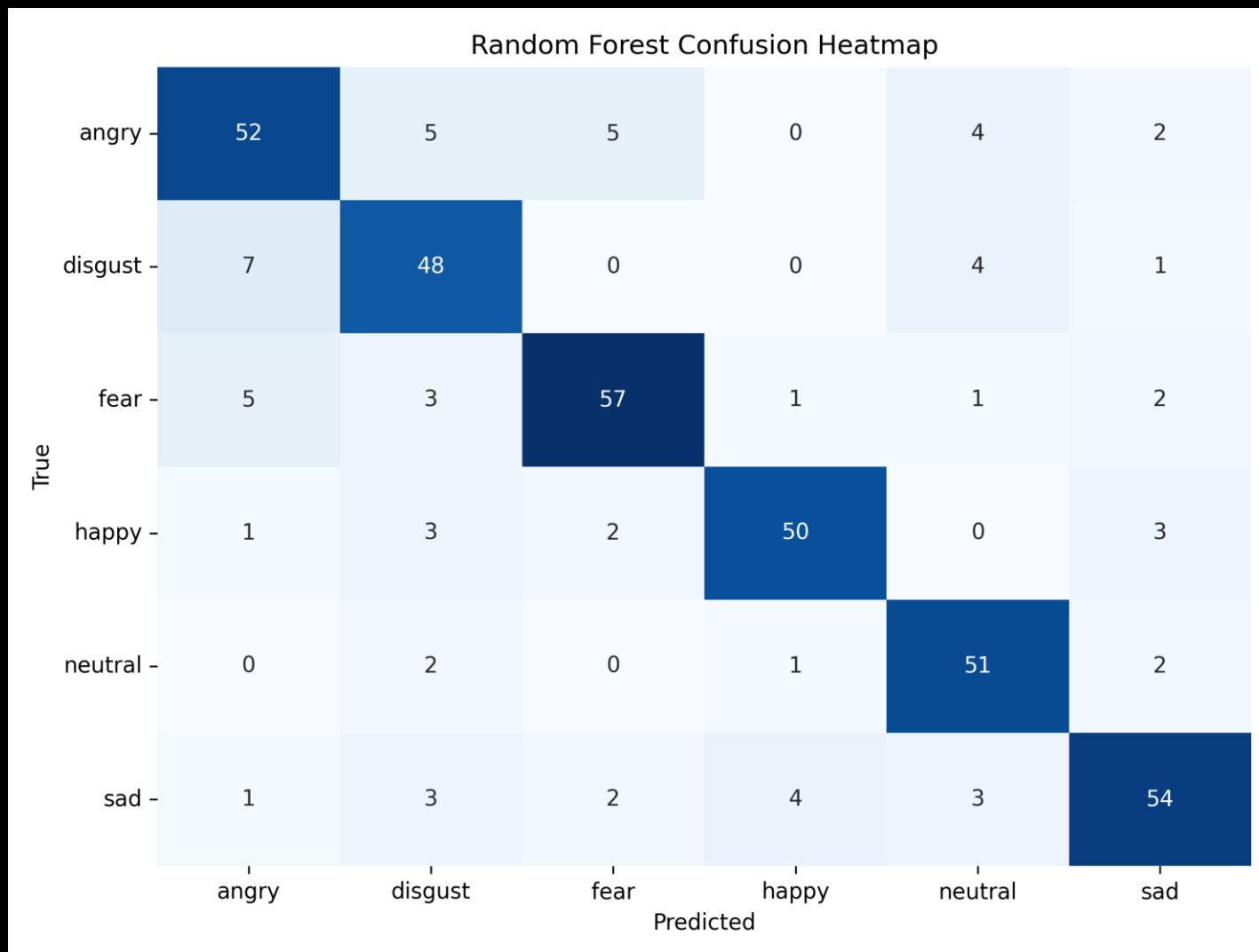
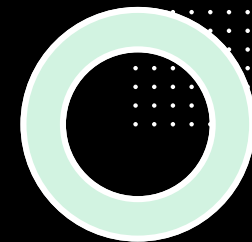
Test Accuracy = 82.98%
Test F1 score = 82.98%

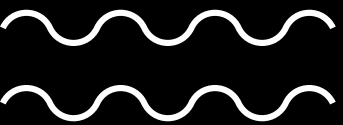




Random Forest

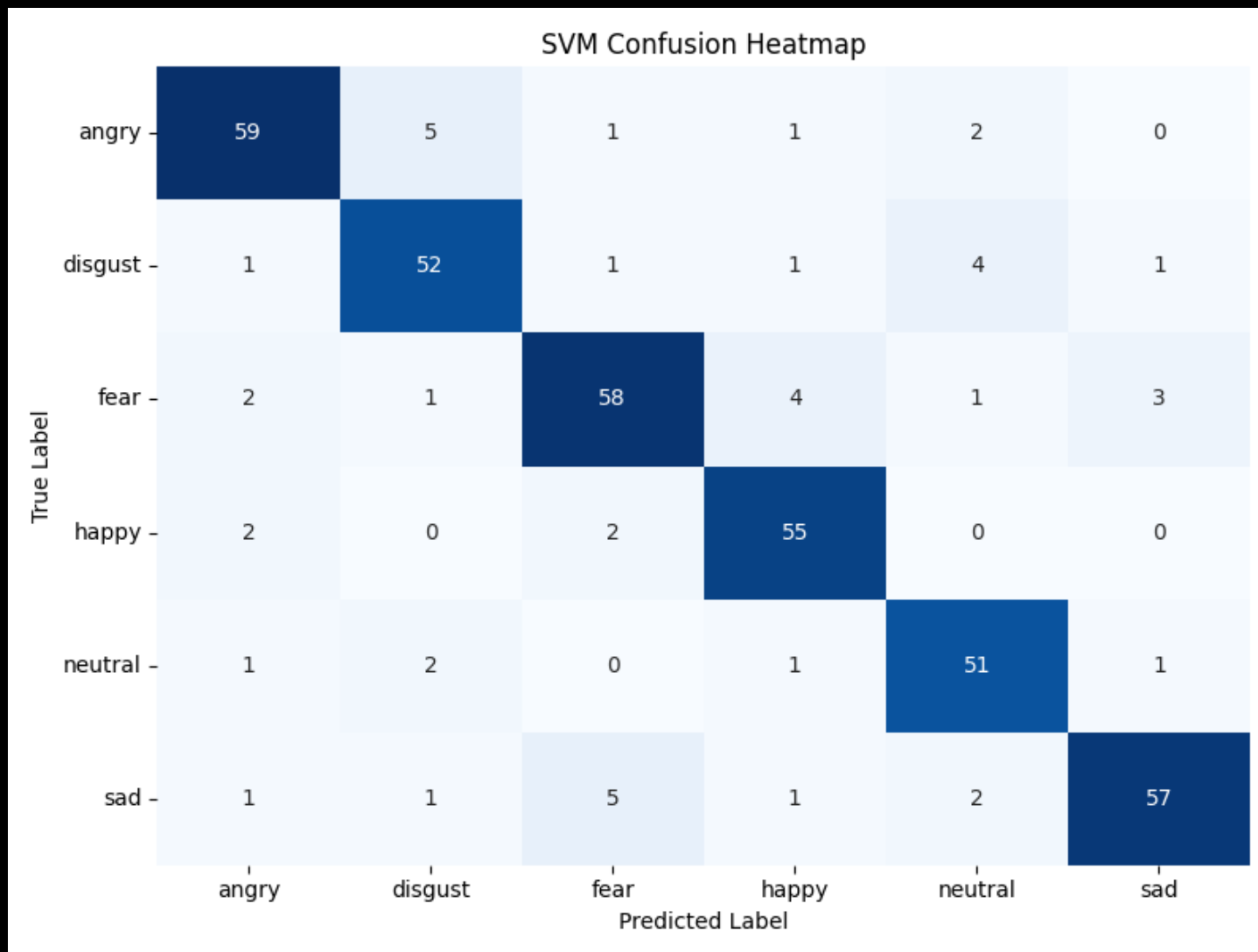
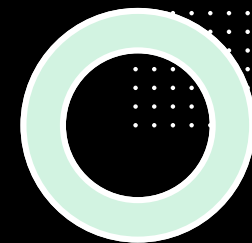
Test Accuracy = 83.25%
Test F1 score = 83.27%

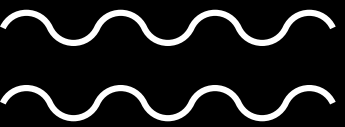




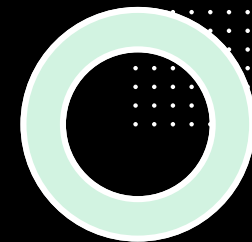
SVM

Test accuracy = 85.86%
Test F1 score = 85.92%



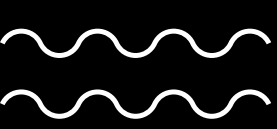


Test Results

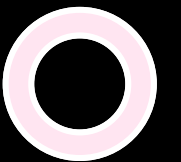


	CNN	MLP	Random Forest	SVM	XGboost
Accuracy	77.75	85.6	83.25	85.86	82.98
F1 score	77.92	85.59	83.27	85.92	82.98





Angry 



THANK
YOU

